QP. Code: 42202

Reg. No. | | | | | | | | |

Second Semester B.C.A.(SEP) Degree Examination, May/June 2025

(Regular)

Numerical and Statistical Methods

Time : 3 Hrs **Max. Marks : 80**

Instructions to Candidates:

1. All the sections are compulsory.
2. Draw neat diagrams wherever necessary.
3. Scientific calculators are allowed.

SECTION - A

Answer any TEN questions. **(10x2=20)**

1. Define transcendental equation. Give an example.
2. Define eigen value and eigen vector.
3. Write a formula to find root of an equation using Seccant - Method.
4. For which system of equations Gauss - Seidel method is applicable?
5. What is an upper triangular matrix?
6. List the application of statistics in business.
7. State the general rules for constructing diagrams and graphs.
8. Explain any four parts of table.
9. Define arithmetic mean and also write down the types of mean.
10. Define co-efficient of Variation.
11. Define ogives or cumulative frequency curves.
12. Define Probability with examples.

SECTION - B

Answer any FOUR questions. **(4x5=20)**

13. By using Bisection Method find the root of an equation $x^3 - 4x - 9 = 0$. What lies between 2 and 3, carry out 3 iterations.
14. Solve the Gauss elimination method.

$$2x + y + z = 0$$
$$3x + 2y + 3z = 18$$
$$x + 4y + 9z = 16$$

15. Distinguish between primary data and secondary data and also discuss various method of collecting primary data.

16. Calculate the median for the following data:

Profits (Rs. in lakhs)	30-32	32-34	34-36	36-38	38-40	40-42	42-44
No. of camp	02	09	25	30	49	62	39

17. If you Pull 2 cards out of a deck, what is the probability that both are spades?

SECTION - C

Answer any FOUR questions. **(4x10=40)**

18. a) Explain the Newton - Ruphson Method to find roots of an equation. **(05)**

    b) Find the root of an equation $x - e^x = \cos x$ by using Regula - falsi method. Carry out 3 iterations. **(05)**

19. a) Using LU - decomposition Method. Solve the following system of equation. **(05)**

    $x + y + z = 1,\ 3x + y - 3z = 5,\ x - 2y - 5z = 10$

    b) Determine the largest eigen value of a matrix. **(05)**

$$\begin{bmatrix} 1 & 6 & 1 \\ 1 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

20. a) For the following frequency distribution of height of students, draw Histogram. **(05)**

Heights (in cms) :	140-145	145-150	150-155	155-160	160-165	165-170
No. of students :	4	10	18	20	19	6

    b) Explain the Pictographs and Cartographs. **(05)**

21. a) Calculate the Quartile deviation and its co-efficient for the following frequency distribution. **(05)**

Daily wages (in Rs.)	3	5	7	9	11	13	15	17	19
No. of persons	2	10	12	15	20	13	12	10	4

    b) Define Mean, Median & Mode with example. **(05)**

22. a) Three coins are tossed at a time. Find **(05)**
    (i) Probability of getting 2 heads
    (ii) Probability of getting no heads
    (iii) Probability of getting 2 heads and 1 tail.

    b) Write down the probability function, range, mean, variance of Bernoulli distribution if [P = 0.2]. **(05)**

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